Year 10 Mathematics

Sets 1

9.5 Shading Sets

Shading Sets

STARTER · 5 MIN

Date: _____ **Title:** Shading Sets

LEARNING OBJECTIVES

- Shade regions for unions and intersections
- Shade regions involving complements
- Describe shaded regions using set notation

STARTER *Answer in your book. No calculators.*

1. What does $A \cap B$ mean?
2. What does $A \cup B$ mean?
3. What does A' mean?
4. Which numbers from 1 to 6 are not in {1, 2, 3}?

Starter — Answers

STARTER

Tick or correct in green pen

1. **In A and in B**
2. **In A or B (or both)**
3. **Not in A**
4. **4, 5, 6**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can shade regions for unions and intersections	☐
SILVER	I can shade regions involving complements	☐
GOLD	I can describe shaded regions using set notation	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Shading Sets

INTRODUCTION · 3 MIN

Shade the region of the Venn diagram that matches the set notation.

$A \cap B$

Shade only the overlap.

AND

$A \cup B$

Shade both whole circles.

OR

A'

Shade everything outside A.

NOT

Harder ones

Shade each part, then combine.

$A' \cap B$: not A, and in B

Examples 1 and 2 — Shading Two Sets

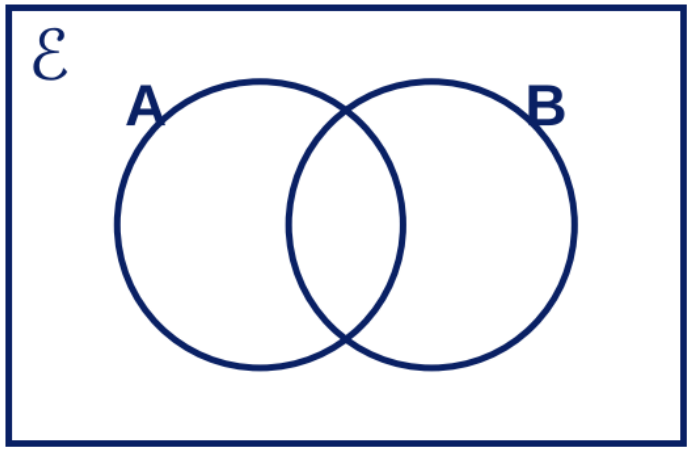
EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

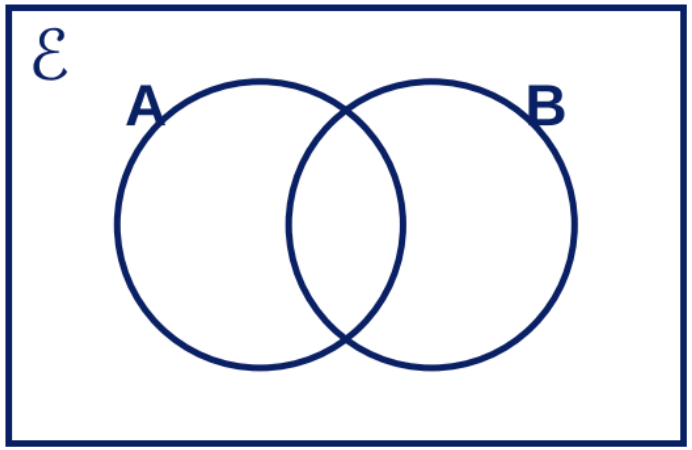
Example 1

Shade $A \cap B$.



Example 2

Shade $A \cup B$.



Examples 1 and 2 — Corrections

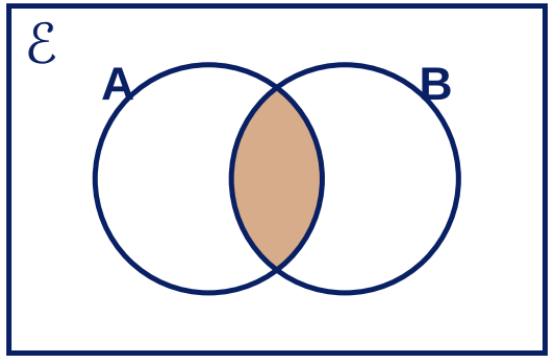
CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

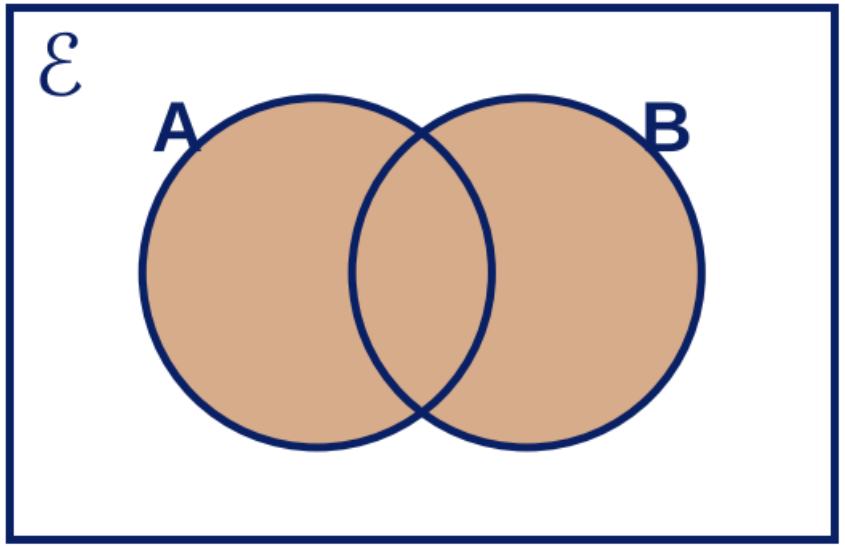
Only the overlap



Tip: ∩ means in both.

Example 2

Both whole circles



Tip: ∪ means in either.

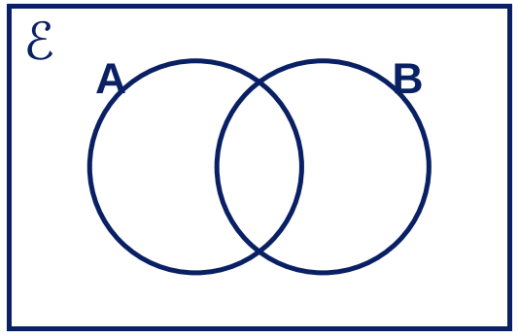
Task 1

PRACTICE · 9 MIN

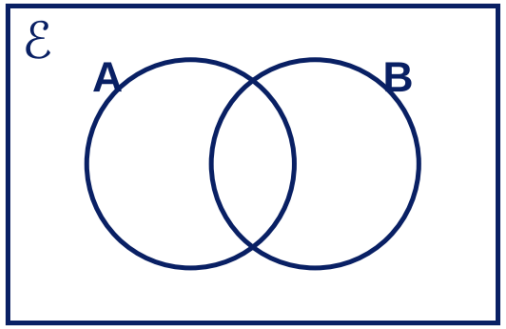
BRONZE

Copy each diagram and shade the set

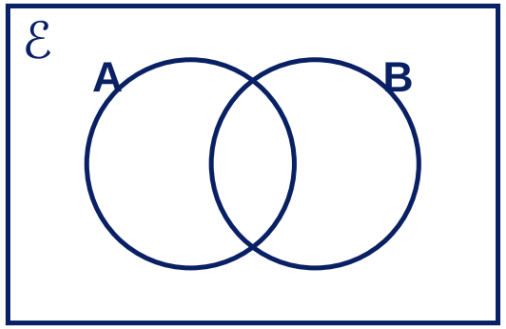
1 A



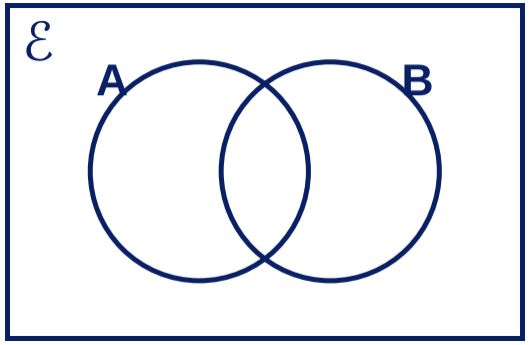
2 B'



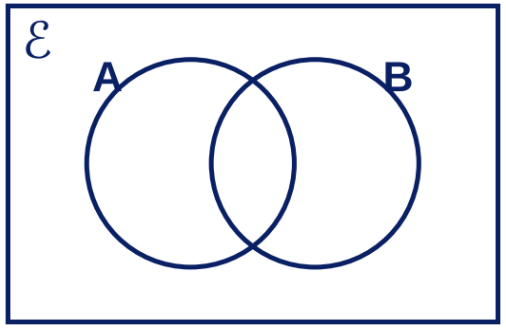
3 A'



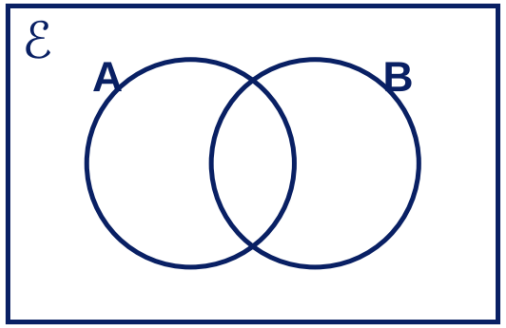
4 $A \cup B$



5 $(A \cup B)'$



★ $A \cap B'$



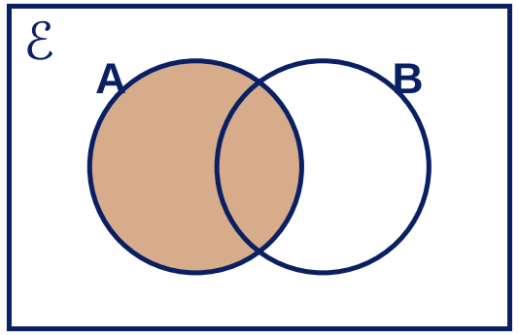
Task 1 — Answers

CORRECTIONS

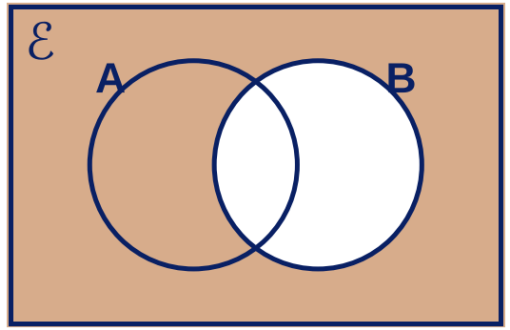
BRONZE

Tick or correct in green pen

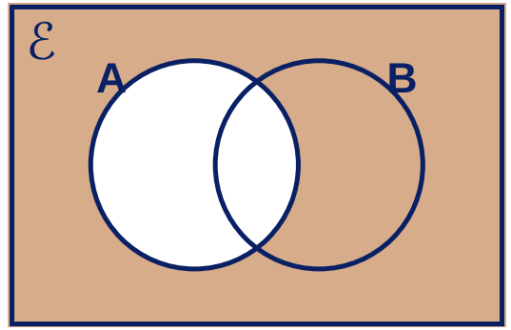
1 A



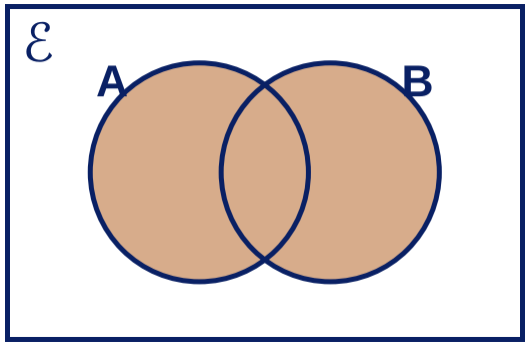
2 B'



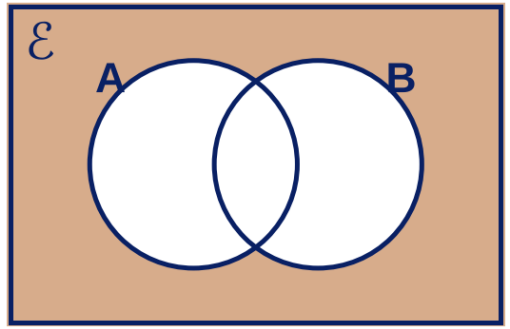
3 A'



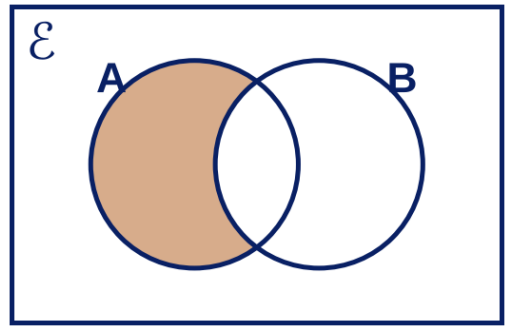
4 $A \cup B$



5 $(A \cup B)'$



★ $A \cap B'$



Examples 3 and 4 — Describing Shaded Regions

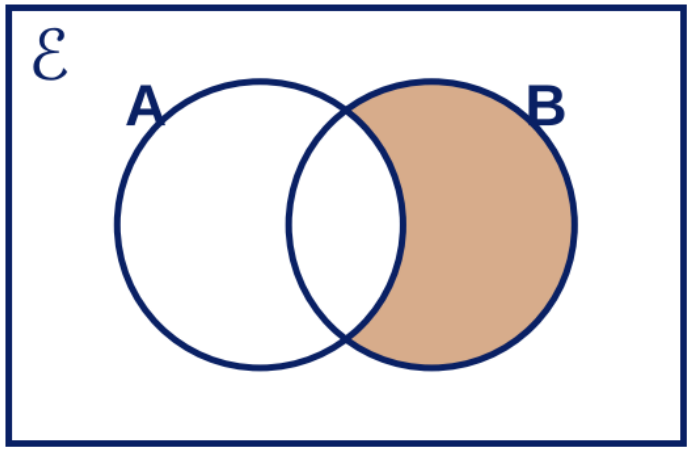
EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

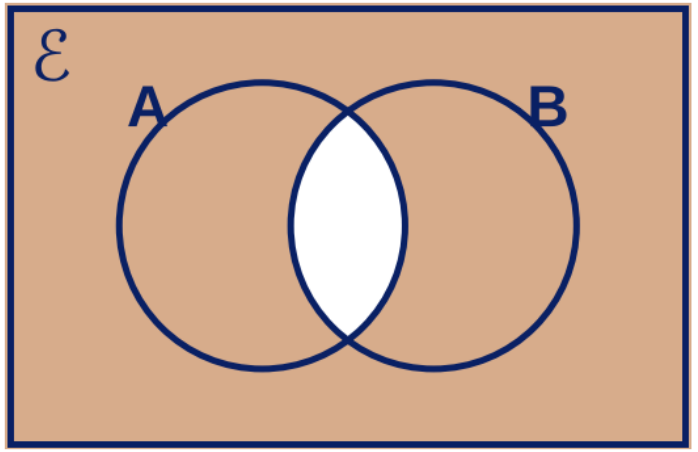
Example 3

Write the shaded region in set notation.



Example 4

Write the shaded region in set notation.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

In B but not in A

$A' \cap B$

Tip: describe it in words first.

Example 4

Everything except the overlap

$(A \cap B)'$

Tip: look at what is NOT shaded.

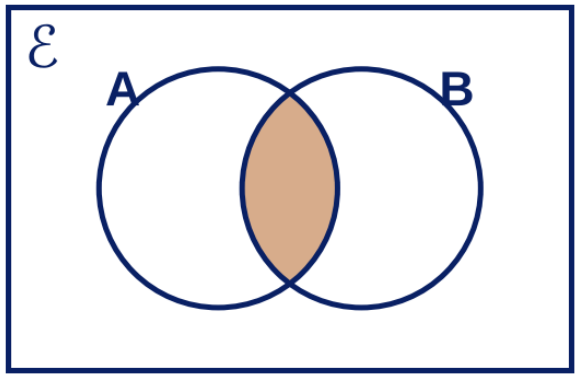
Task 2

PRACTICE · 9 MIN

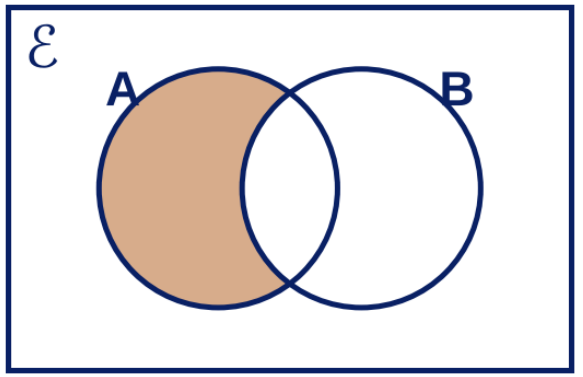
SILVER

Write each shaded region in set notation

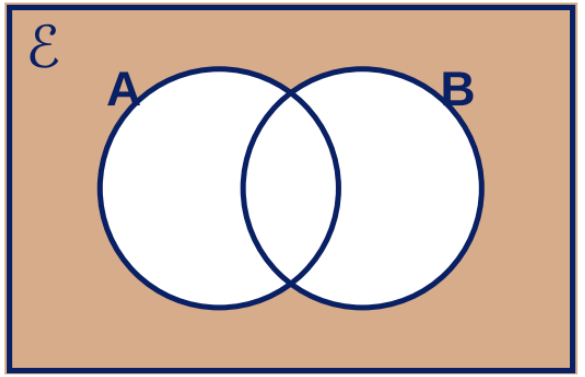
1



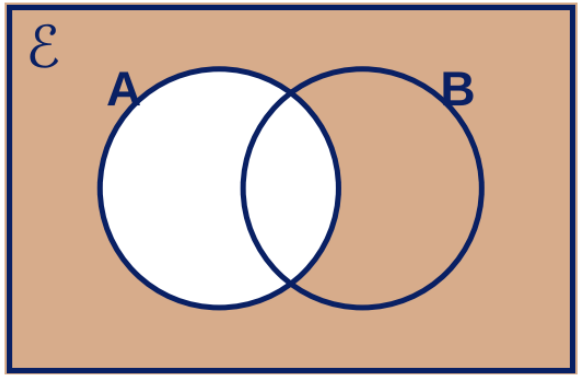
2



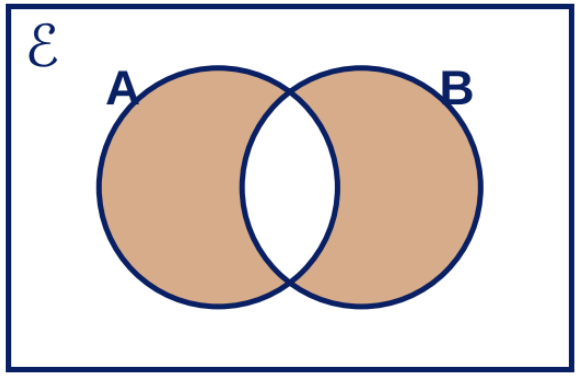
3



4



5



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $A \cap B$
2. $A \cap B'$
3. $(A \cup B)'$
4. A'
5. $(A \cup B) \cap (A \cap B)'$

Examples 5 and 6 — Shading Three Sets

EXAMPLES · 5 MIN

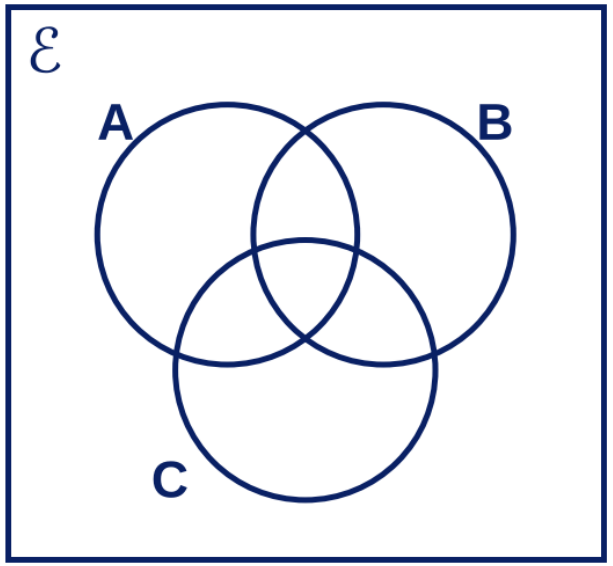
GOLD

Exam-style: show all your working

Example 5

(1 mark)

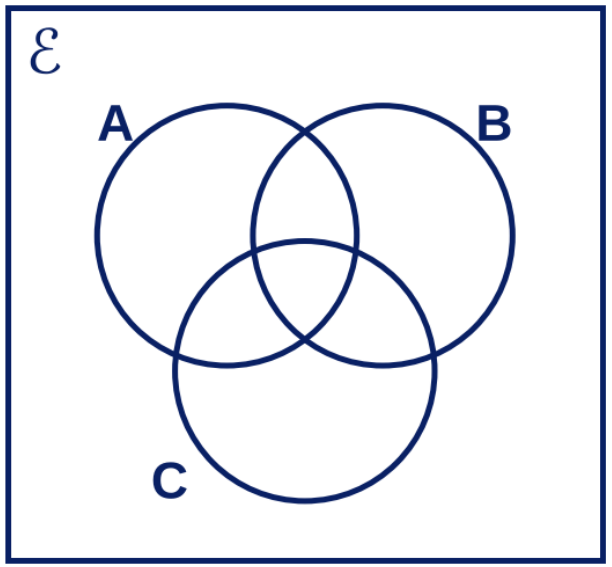
Shade $A \cap B \cap C$.



Example 6

(2 marks)

Shade $(A \cup B) \cap C$.



Examples 5 and 6 — Corrections

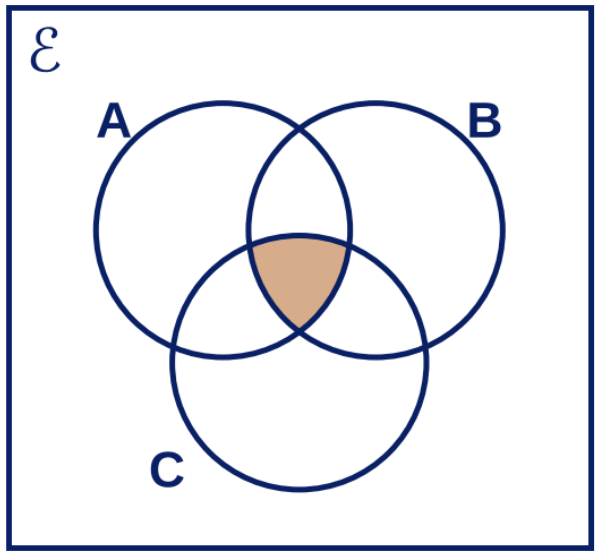
CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

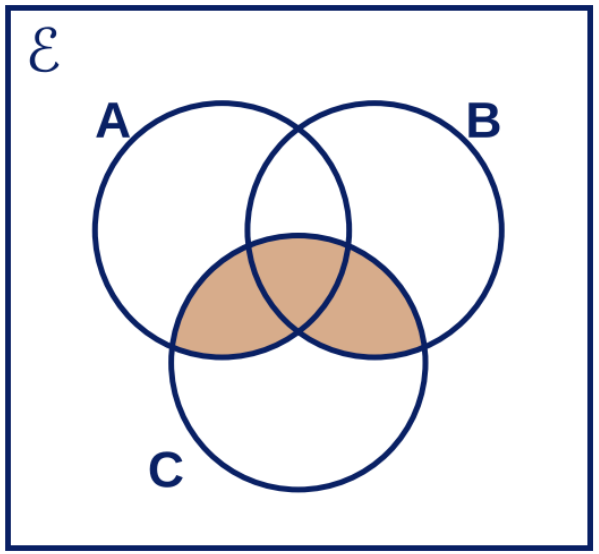
Only the centre region



Tip: in all three circles.

Example 6

Parts of C that are also in A or B



Tip: shade A ∪ B, then keep only the part in C.

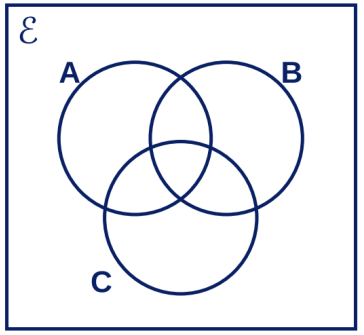
Task 3

PRACTICE · 9 MIN

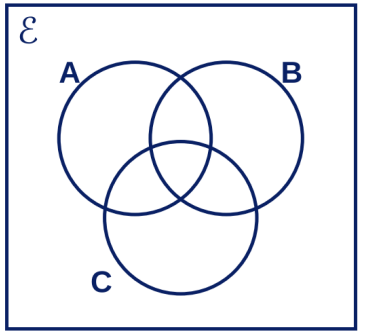
GOLD

Copy each diagram and shade the set

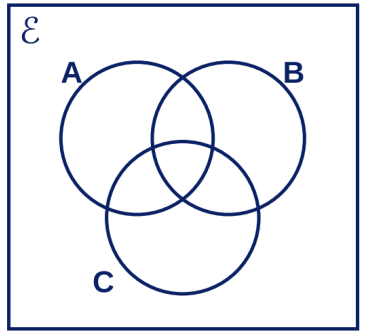
1 C'



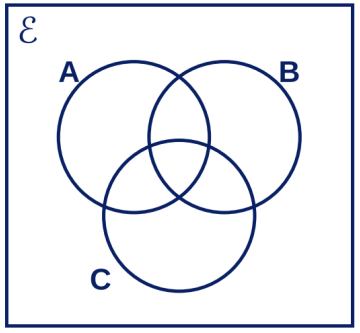
2 $A \cap B$



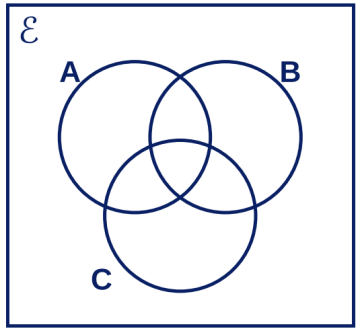
3 $A \cup C$



4 $(A \cup B \cup C)'$



5 $A \cap B \cap C'$



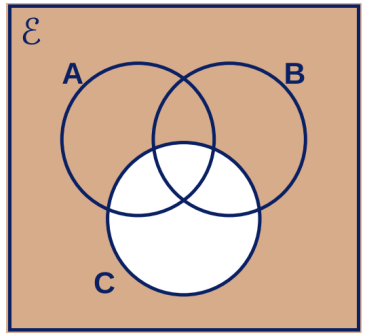
Task 3 — Answers

CORRECTIONS

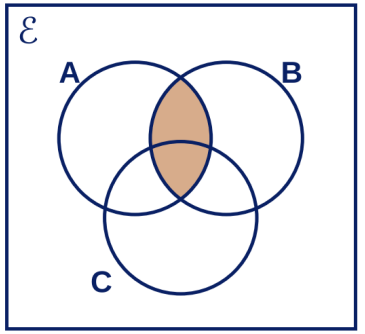
GOLD

Tick or correct in green pen

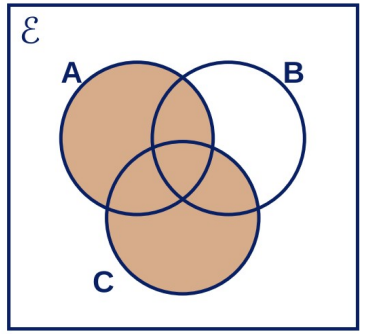
1 C'



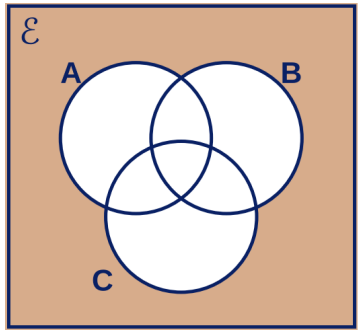
2 $A \cap B$



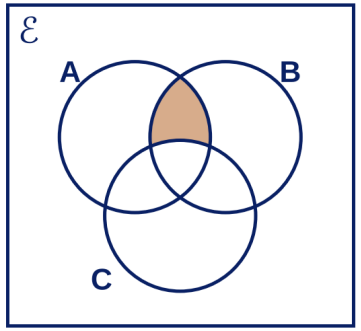
3 $A \cup C$



4 $(A \cup B \cup C)'$



5 $A \cap B \cap C'$



Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Which region is $A \cap B$?

SILVER Which region is A'?

GOLD Write the set notation for 'in B but not in A'.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **The overlap of A and B**

SILVER **Everything outside circle A**

GOLD **$A' \cap B$**

Exam Practice — Shading Two Sets

EXAM PRACTICE · 10 MIN

BRONZE

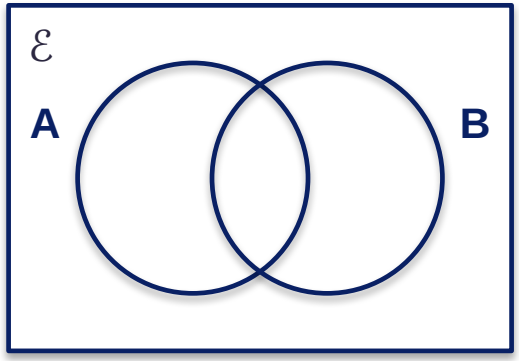
IGCSE exam-style: answer on the line, show all your working

Question 1

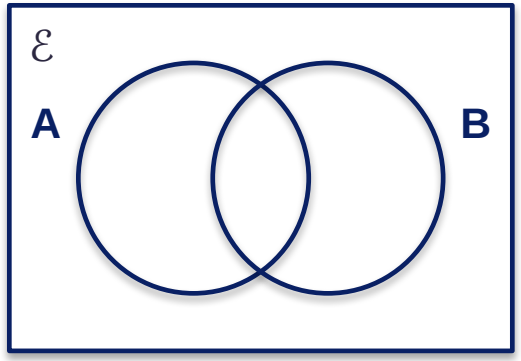
(2 marks)

Shade each set on its Venn diagram.

(a) $A \cap B$



(b) $A \cup B$

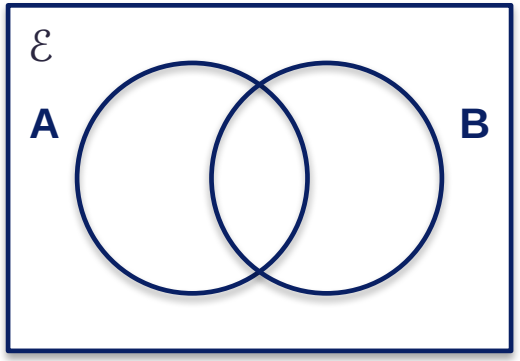


Question 2

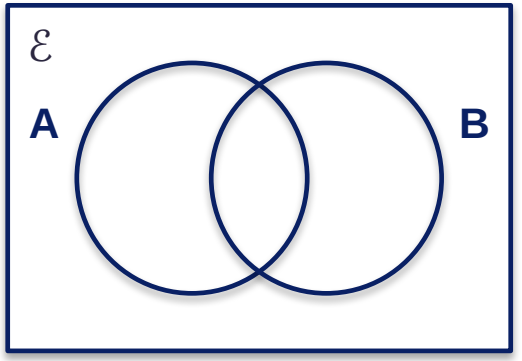
(2 marks)

Shade each set on its Venn diagram.

(a) A'



(b) $(A \cup B)'$



Exam Practice — Describing Shaded Regions

EXAM PRACTICE · 10 MIN

SILVER

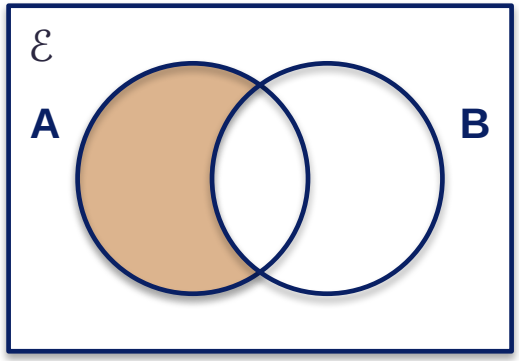
IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

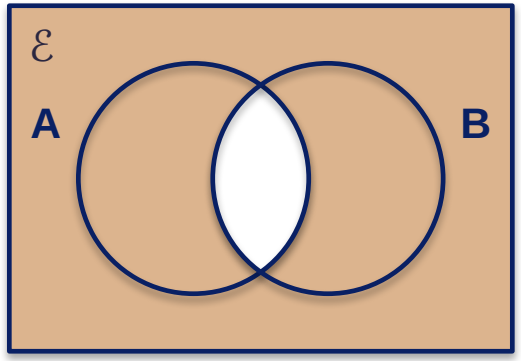
Use set notation to describe each shaded region.

(a)



.....

(b)



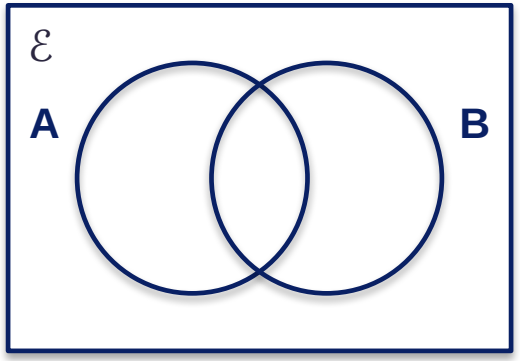
.....

Question 4

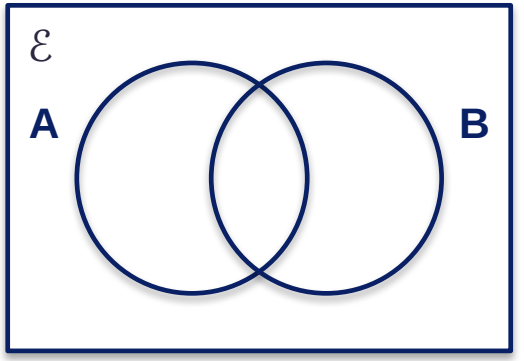
(2 marks)

Shade each set on its Venn diagram.

(a) $A' \cap B$



(b) $A \cup B'$



Exam Practice — Shading Three Sets

EXAM PRACTICE · 10 MIN

GOLD

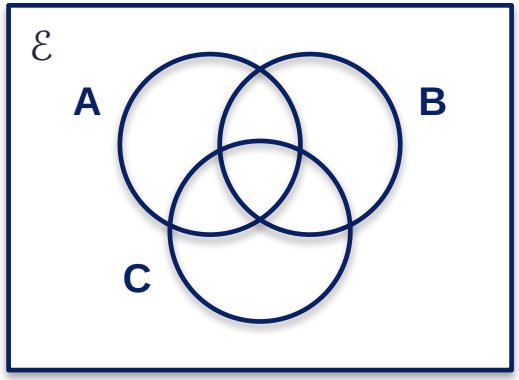
IGCSE exam-style: answer on the line, show all your working

Question 5

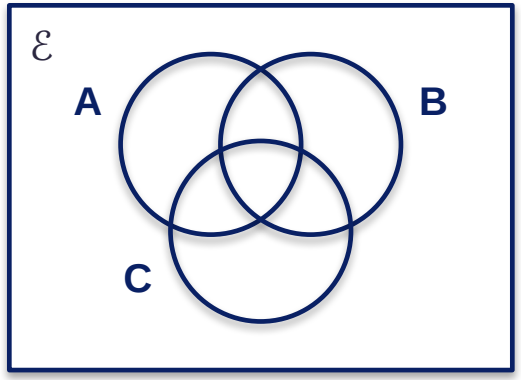
(2 marks)

Shade each set on its Venn diagram.

(a) $A \cap B \cap C'$



(b) $A \cup (B \cap C)$



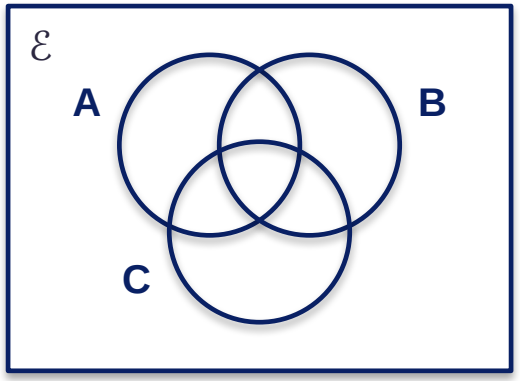
Question 6

(2 marks)

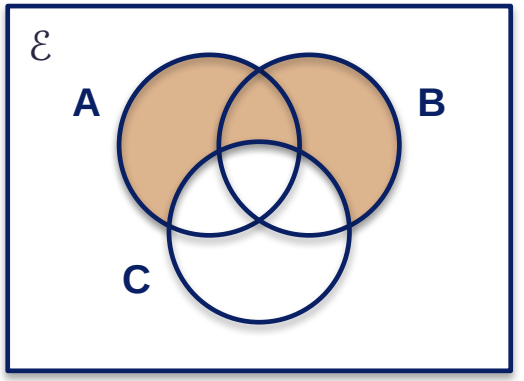
(a) Shade $(A \cap B)' \cap C$.

(b) Describe the shaded region.

(a)



(b)



.....

Exam Practice — Mark Scheme

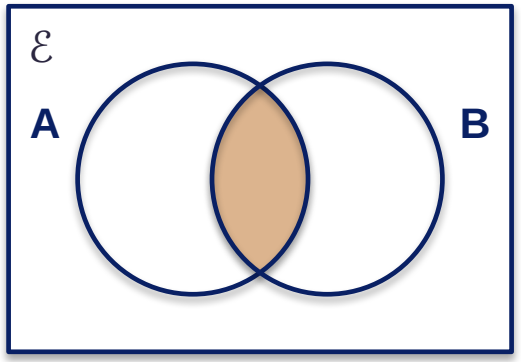
CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

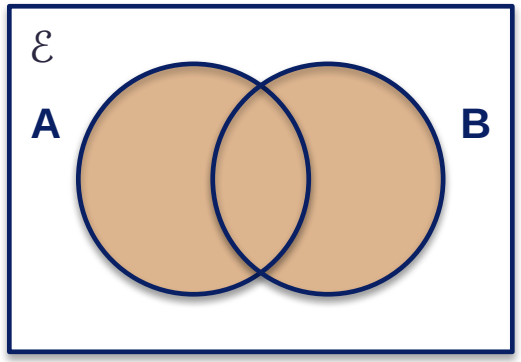
Question 1

(a) $A \cap B$



B1

(b) $A \cup B$

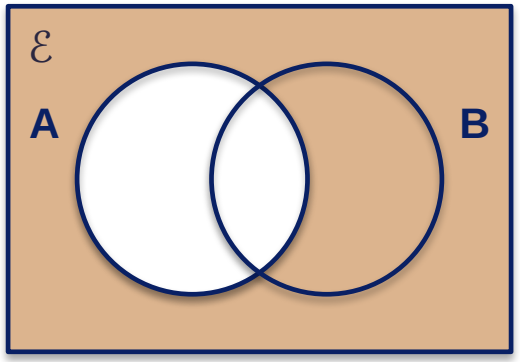


B1

Tip: ∩ means in both, ∪ means in either.

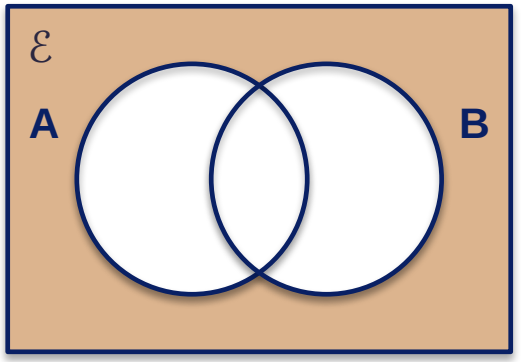
Question 2

(a) A'



B1

(b) $(A \cup B)'$



B1

Tip: ′ means not in, so shade outside.

Exam Practice — Mark Scheme

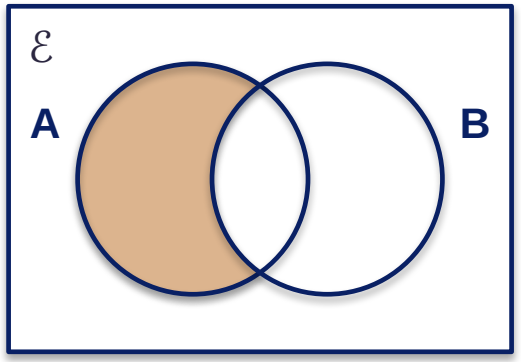
CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

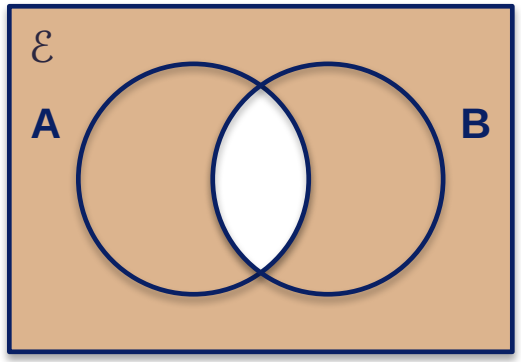
(a)



A ∩ B′

B1

(b)



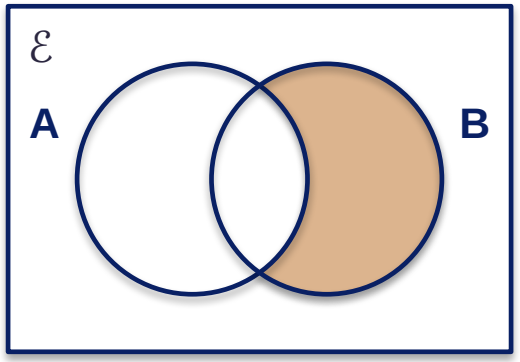
(A ∩ B)′

B1

Tip: say it in words first: in A, not in B.

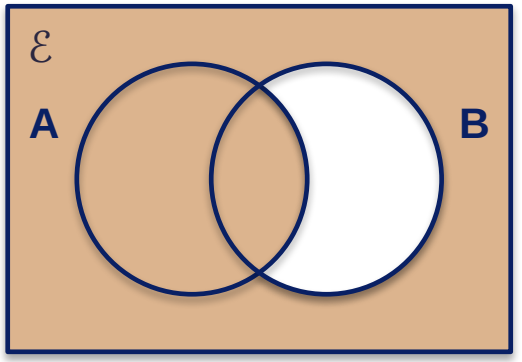
Question 4

(a) A′ ∩ B



B1

(b) A ∪ B′



B1

Tip: shade B′ and A, then keep both.

Exam Practice — Mark Scheme

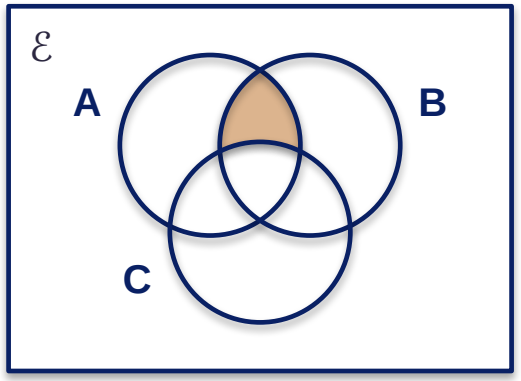
CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

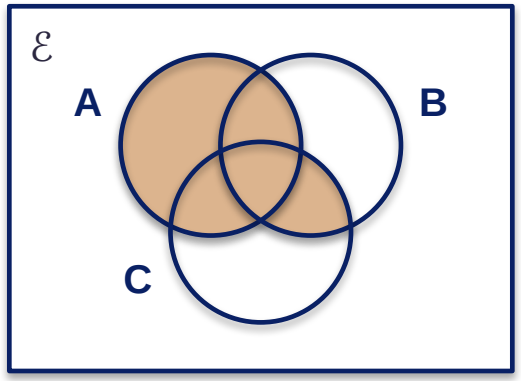
Question 5

(a) $A \cap B \cap C'$



B1

(b) $A \cup (B \cap C)$

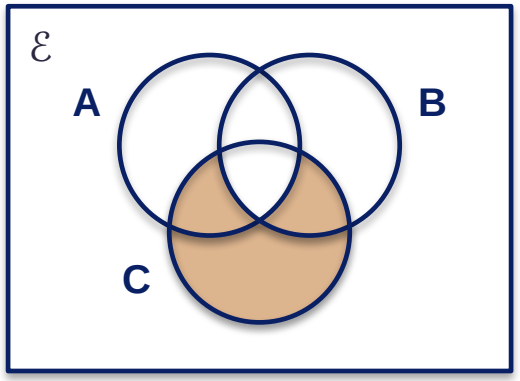


B1

Tip: shade the bracket first, then combine.

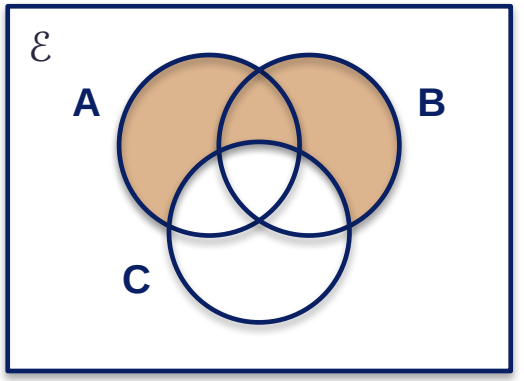
Question 6

(a)



B1

(b)



$(A \cup B) \cap C'$

B1

Tip: (b) also accept $C' \cap (A \cup B)$.